Multi-Objective Optimization for Fairness in Income Prediction

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1 Problem Statement

We address the trade-off between predictive performance and algorithmic fairness in income prediction. The task is to predict whether an individual earns more than \$50,000 per year using the Adult Census Income dataset [1], while simultaneously optimising for fairness with respect to gender as the protected attribute.

The dataset contains 30,162 samples after cleaning, with 87 features after one-hot encoding, and a class imbalance of approximately 75% (income $\leq$\$50K) vs. 25% (income >\$50K). The sensitive attributes `sex` and `race` are retained as features in the model. Removing them does not eliminate bias because proxy features such as `occupation` and `relationship` are strongly correlated with gender. Retaining sensitive attributes and imposing fairness as an explicit objective is more transparent and produces a more meaningful Pareto front.

2 Formulation

2.1 Decision Variables

The search space is modelled continuously, with variables cast to integers or categorical choices during evaluation. Each candidate solution is a vector $\mathbf{x} \in R^6$:

Index	Variable	Continuous Range	Interpretation
x_0	Model type	$\{0, 1, 2\}$	Selects the learning algorithm
x_1	Regularisation	$[0.001, 100.0]$	Controls penalty strength to prevent overfitting
x_2	Max depth	$[2, 15]$	Limits tree depth
x_3	No. of estimators	$[50, 200]$	Sets number of trees in the ensemble
x_4	Decision Threshold	$[0.3, 0.7]$	Cutoff for +ve outcomes
x_5	Imbalance strategy	$\{0, 1\}$	Class imbalance (none or weighting)

2.2 Objectives

All three objectives are cast as minimisation problems:

$$f_1(\mathbf{x}) = -\,\text{BalancedAccuracy}(y, \hat{y}) \tag{1}$$

$$f_2(\mathbf{x}) = |P(\hat{y} = 1 \mid G = \text{Male}) - P(\hat{y} = 1 \mid G = \text{Female})| \tag{2}$$

$$f_3(\mathbf{x}) = \max(|\text{TPR}_{\text{Male}} - \text{TPR}_{\text{Female}}|,\ |\text{FPR}_{\text{Male}} - \text{FPR}_{\text{Female}}|) \tag{3}$$

f_2 is the Demographic Parity Difference (DPD) and f_3 is the Equalized Odds Difference (EOD), computed using `fairlearn`. Balanced accuracy handles the 75/25 class imbalance.

3 Algorithm

We use NSGA-III implemented in `pymoo` [2]. NSGA-III was chosen because its structured reference directions (Das-Dennis method) enforce uniform coverage of the Pareto surface across three objectives better than NSGA-II's crowding distance.

Configuration:

- Reference directions: Das-Dennis, $n_\text{partitions} = 10 \rightarrow 66$ reference points
- Population size: 66
- Sampling: Continuous Float Random Sampling
- Crossover: SBX, $p = 0.9$, $\eta = 5$
- Mutation: Polynomial mutation, $p = 0.4$, $\eta = 5$
- Generations: 20, Total evaluations: $\sim$1320

4 Experimental Setup

We performed three distinct multi-objective optimization runs to isolate the impact of the class imbalance strategy: one restricted to `None`, one restricted to `class_weight`, and a combined run (`both`) where the optimizer selected the strategy dynamically.

5 Quantitative Analysis

5.1 Pareto Front Visualisation

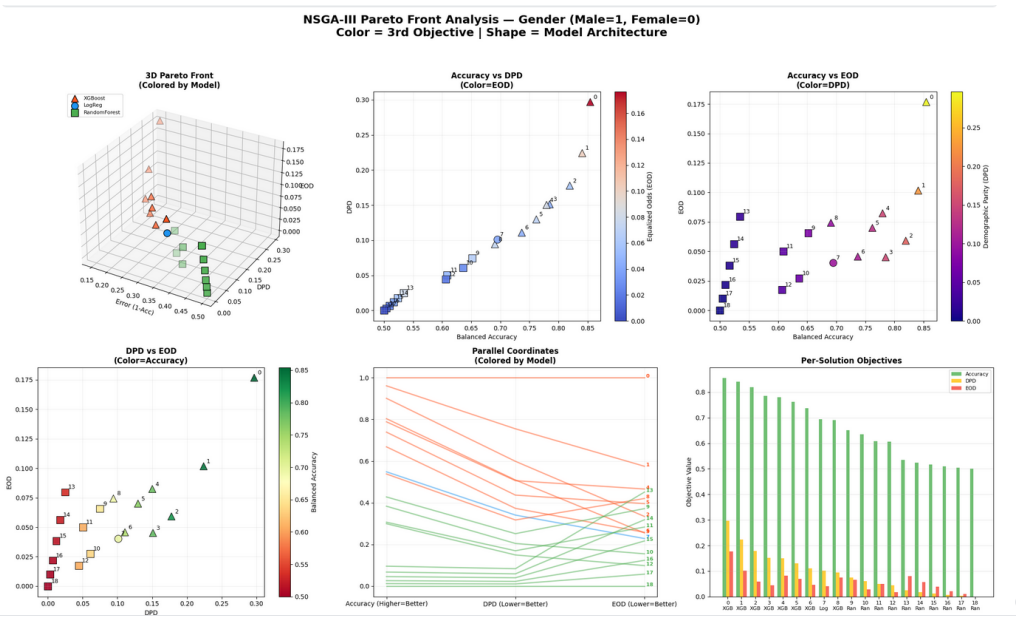

Figure 1: NSGA-III Pareto front for the combined strategy run. Includes 3D scatter, Accuracy vs Fairness 2D projections, and hyperparameter distributions.

5.2 Hypervolume and Spread Metric

To objectively compare the quality of the Pareto fronts generated by different class imbalance strategies, we computed the Hypervolume Indicator (HV) and the Spacing Metric (S).

Reference point for Hypervolume: $\mathbf{r} = (0.0,\ 0.35,\ 0.35)$. This represents a baseline of zero accuracy gain, and maximum acceptable bounds of 0.35 for both DPD and EOD.

Table 1: Quantitative comparison of NSGA-III runs across different imbalance strategies.

Imbalance Strategy Allowed	Hypervolume (HV) ↑	Spacing Metric (S) ↓
Only `None`	**0.089553**	**0.024185**
Both Allowed	0.088526	0.034296
Only `class_weight`	0.061959	0.046993

Analysis of Metrics: The quantitative metrics strictly reinforce the inferiority of algorithmic class weighting in this multi-objective context. The run restricted to `None` achieved a **44% higher Hypervolume** than the `class_weight` run (0.0895 vs. 0.0619). This indicates that the `None` strategy dominates a vastly superior volume of the objective space, pushing much further into the high-accuracy/low-bias regions.

Furthermore, the Spacing Metric reveals that the `None` strategy produced a much more uniform and evenly distributed Pareto front ($S = 0.024$). In contrast, forcing the optimizer to use `class_weight` resulted in a highly fragmented front ($S = 0.046$), suffering from large gaps and poor trade-off continuity. When the optimizer was allowed to choose between both strategies, it essentially approximated the `None` front, relying on `class_weight` for only one extreme high-accuracy outlier, which slightly degraded its overall spacing score.

5.3 Pareto Points Tabulation

Table 2 displays a diverse, representative subset of the 19 non-dominated solutions discovered when the optimizer was permitted to select between 'None' and 'class_weight' dynamically.

Table 2: Selected Pareto-optimal solutions from the Combined NSGA-III run.

Idx	Model	Reg	Depth	N-Est	Thresh	Imbalance	Bal.Acc	DPD / EOD
0	XGBoost	92.31	8	123	0.49	class_weight	0.8546	0.2968 / 0.1767
1	XGBoost	77.79	8	120	0.36	None	0.8407	0.2240 / 0.1016
2	XGBoost	19.071	6	158	0.453	None	0.82	0.177 / 0.059
6	XGBoost	13.41	7	136	0.69	None	0.734	0.11 / 0.045
7	LogReg	12.53	13	84	0.70	None	0.6945	0.1012 / 0.0404
12	RandomForest	65.88	8	64	0.68	None	0.6062	0.0444 / 0.0174
13	RandomForest	66.07	4	153	0.67	None	0.5340	0.0249 / 0.0797
18	RandomForest	28.88	3	200	0.68	None	0.5000	0.0000 / 0.0000

Table 3 displays selected non-dominated solutions discovered when the optimizer was forced to use the None strategy. The threshold hyperparameter may already account for the class imbalance although via a post-hoc intervention. Thus, it makes sense to look at these solutions. In the class_weight strategy we get lesser pareto points most of which are dominated by the None strategy. Hence we don't present those points here.

Objective Ranges (Combined Run):

- **Balanced Accuracy:** $0.5000 \rightarrow 0.8546$
- **DPD:** $0.0000 \rightarrow 0.2968$
- **EOD:** $0.0000 \rightarrow 0.1767$

Table 3: Selected pareto optimal solutions using the None strategy.

Idx	Model	Reg	Depth	N-Est	Thresh	Imbalance	Bal.Acc	DPD / EOD
0	XGBoost	80.44	7	167	0.31	None	0.8487	0.2587 / 0.1381
1	XGBoost	12.86	8	134	0.40	None	0.8318	0.2014 / 0.0810
2	XGBoost	28.45	11	80	0.46	None	0.8156	0.1757 / 0.0592
8	LogReg	2.72	15	53	0.68	None	0.7039	0.1098 / 0.0458
16	RandomForest	2.29	4	187	0.62	None	0.5486	0.0295 / 0.0718
17	RandomForest	21.40	4	89	0.69	None	0.5120	0.0088 / 0.0281
18	RandomForest	34.84	3	148	0.70	None	0.5000	0.0000 / 0.0000

6 Qualitative Analysis

6.1 Core Accuracy vs. Demographic Parity Trade-off

From both tables we see that across all models, improving predictive performance significantly worsens demographic parity. For instance, moving from a relatively balanced Logistic Regression model (Solution 7: Accuracy 0.694, DPD 0.101) to a highly accurate XGBoost model (Solution 1: Accuracy 0.840, DPD 0.224) increases accuracy by 14.6 percentage points, but more than doubles the demographic disparity (a 121% relative increase). This highlights a strong inverse relationship between accuracy and group fairness.
XGBoost solutions tend more towards optimizing accuracy while Random Forests focus on fairness. Logisitic Regression gives a single non-dominated point in the middle.

6.2 Tree Complexity vs. Strict Fairness

Constraining model complexity enforces fairness at the cost of utility. A shallow Random Forest (Solution 18, max_depth=3) achieves perfect fairness (DPD=0, EOD=0) but performs no better than random guessing (Accuracy $\approx$ 0.500). In contrast, increasing depth (Solution 1, max_depth=8) allows the model to capture richer feature interactions, improving accuracy to 0.840, but introduces a demographic parity gap of 0.224. This demonstrates that model expressiveness directly contributes to fairness violations.

6.3 Practically Useful Solutions

If fairness is not that much of a concern then **Solution 1** with **balanced_acc=0.841, DPD=0.224 , EOD=0.102** gives a good accuracy staying well over 80 percent while also giving a decent EOD value.
If fairness is of greater importance then **Solution 6** with **balanced_acc=0.737, DPD=0.111, EOD=0.046** is a decent choice as it keeps DPD and EOD low while maintaining a reasonable accuracy of over 70 percent.

6.4 Strengths and Weaknesses of NSGA-III

Strengths: Structured Many-Objective Scaling
NSGA-III is uniquely suited for this formulation because it was explicitly designed to scale beyond two objectives. Rather than relying on crowding distance algorithms that collapse in 3D space, NSGA-III projects a uniform grid of reference directions (the Das-Dennis method) to anchor the population. This guarantees a highly uniform and evenly distributed Pareto surface across Accuracy, DPD, and EOD, ensuring that no single objective dominates the evolutionary search.

Weaknesses: Utility Blindness and Sample Inefficiency
Like all pure MOO algorithms, NSGA-III is "practically blind"—it optimizes purely for mathe-

matical non-dominance without context. Consequently, it expends significant computational effort isolating perfectly fair but practically useless models (e.g., accuracy collapsing to a random-guess baseline). Furthermore, uniquely compared to Bayesian approaches (e.g., BoTorch with qEHVI), NSGA-III is entirely sample-inefficient. It relies on brute-force genetic mutation without building a predictive surrogate model of the search space. It is only viable here because evaluating a candidate model is computationally cheap ($\sim$10 seconds); in a deep-learning context, its reliance on hundreds of exploratory evaluations would be prohibitively expensive.

6.5 Macro-Level Correlation Between DPD and EOD

Across the Pareto front, DPD and EOD generally exhibit a positive correlation at the macro level. As models achieve fairer positive prediction rates across groups, disparities in EOD also tend to decrease.

For example, descending from the high-accuracy XGBoost model (Solution 1: DPD = 0.224, EOD = 0.102) to a more balanced Logistic Regression model (Solution 7: DPD = 0.101, EOD = 0.040), both fairness gaps are reduced by approximately half simultaneously.

However, as the solutions push toward near-zero bias at the extreme edge of the Pareto front, this cooperative relationship collapses, revealing a severe and non-monotonic conflict between the two fairness objectives.

6.6 A Non-Obvious Tradeoff

A non-obvious finding emerges when analyzing this conflict more closely. Intuitively, one might expect fairness metrics to be positively correlated—that as a model becomes "fairer" overall. However, the Pareto front reveals that DPD and EOD can conflict non-monotonically. This phenomenon is clearly illustrated by comparing two Random Forest solutions on the Pareto front:

- **Solution 12** achieves a highly balanced state with DPD = 0.044 and EOD = 0.017.
- **Solution 13** pushes for stricter demographic parity, improving DPD to 0.025. However, this comes at a severe cost: EOD explodes by over 360%, rising to 0.079.

This counter-intuitive behavior arises because the Adult dataset contains historical ground-truth differences in income between genders (unequal base rates). When the optimizer aggressively forces DPD close to zero—demanding identical positive prediction rates for both men and women—it must artificially boost predictions for women while suppressing them for men.

To achieve this, the model is forced to misclassify high-earning men as low-earning or low-earning women as high-earning. These injected errors severely disrupt the balance of True Positive and False Positive rates across groups, resulting in a sharp spike in Equalized Odds Difference.

6.7 The Impact of Class Imbalance Strategies

By isolating runs with and without class weights, a profound architectural trade-off emerges:

- **The Penalty of Class Weights:** When NSGA-III is restricted to using `class_weight`, EOD spikes dramatically, reaching as high as 0.3755. By forcing the model to heavily weight the minority class, the model becomes hyper-sensitive to features correlated with wealth, which in this dataset may serve as heavily gendered proxies (e.g., specific occupations, hours worked).

- **The Optimizer's Natural Preference:** In the combined run where the algorithm could freely choose the imbalance strategy, *18 out of the 19 Pareto-optimal solutions utilized "None"*. The optimizer only resorted to `class_weight` for Solution 0, which maximized accuracy at the absolute detriment of fairness.

7 Conclusion

NSGA-III effectively mapped the accuracy–fairness frontier over a continuous hyperparameter space. The primary takeaway is that **correcting for class imbalance severely harms Equalized Odds** in income prediction. When left to its own devices, a multi-objective optimizer minimizing algorithmic bias actively discards class-weighting mechanisms in favor of standard training logic, utilizing decision thresholds and shallow tree depths to map out the Pareto surface.

References

[1] Dua, D. and Graff, C. (2019). UCI Machine Learning Repository — Adult Dataset. `https://www.kaggle.com/datasets/uciml/adult-census-income`

[2] Blank, J. and Deb, K. (2020). pymoo: Multi-objective optimisation in Python. *IEEE Access*, 8, 89497–89509.

A Pareto-Optimal Solution Explained in Detail

Solution 2 is a good solution that successfully balances high predictive performance (maintaining $> 80\%$ accuracy) with decent fairness constraints, nearly halving the EOD penalty compared to the most accurate models.

Decision variables:

Variable	Value	Interpretation
Model type (x_0)	2 (XGBoost)	Gradient boosted decision trees
Regularisation (x_1)	19.07	Light L2 Regularization
Max depth (x_2)	6	Shallow depth to prevent overfitting on majority groups
No. of estimators (x_3)	158	High ensemble size for stability
Decision Thresh (x_4)	0.45	Slightly lowered threshold to compensate for imbalance
Imbalance strategy (x_5)	0 (None)	No class reweighting

Objective values:

Objective	Value	Interpretation
Balanced Accuracy	0.8194	Strong predictive performance (well above random/baseline)
DPD	0.1777	17.7% discrepancy in positive prediction rate across groups
EOD	0.0591	True positive rate gap between groups is minimal (under 6%)

This solution sits at the critical "knee" of the Pareto front. Compared to Solution 1, it trades a marginal $\sim$2.1% drop in Balanced Accuracy for a massive $\sim$42% relative reduction in Equalized Odds Difference (EOD). It achieves this by combining a shallower maximum tree depth (6) with a higher number of estimators (158) and light regularization, preventing the model from deeply memorizing biased patterns in the training data while remaining highly accurate.